

**SEMi-Complete by Design: A Monte Carlo simulation to assess Measurement
Invariance in Moderated Nonlinear Factor Analysis and SEM-Trees**

Leonie Hagitte^{1,2,3} and Andreas M. Brandmaier^{2,3,4}

¹Goethe University Frankfurt

²Max Planck Institute for Human Development, Berlin, Germany

³MSB Medical School Berlin, Berlin, Germany

⁴Max Planck UCL Centre for Computational Psychiatry and Ageing Research, Berlin,
Germany

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General Information

What is the title of the project?

SEMi-Complete by Design: A Monte Carlo simulation to assess Measurement Invariance in Moderated Nonlinear Factor Analysis and SEM-Trees.

Who are the current and future project contributors?

Leonie Hagitte, Andreas M. Brandmaier

Provide a description of the project.

Ensuring the validity of psychological assessments is crucial, yet differential item functioning (DIF) can threaten measurement invariance (MI) when test items function differently across groups (Bauer et al. 2020). Recent calls for improved DIF detection methods emphasize the need for more advanced statistical approaches (Lee et al. 2024). Moderated nonlinear factor analysis (MNLFA) is a recent approach for assessing MI via parameter moderation within a single-group confirmatory factor analysis framework (Bauer 2017). MNLFA evaluates MI across multiple continuous and categorical covariates, and accounts for heteroskedasticity by modeling factor and residual variances as functions of these covariates. While MNLFA offers continuous moderation of several parameters of SEMs (e.g.: factor loadings, covariances etc.), it requires a priori specification of covariates and their functional relationships, which are usually assumed to be linear (Bauer 2017; Kolbe et al. 2024). In contrast, structural equation modeling (SEM) trees and forests are data-driven, non-parametric methods that use recursive partitioning to identify latent subgroups in which model parameters differ, without assuming specific functional forms or predefined covariate effects. These approaches allow for nonlinear moderation of factor loadings and can reveal complex interaction effects, enabling the exploratory detection of DIF (Brandmaier et al. 2016, 2013). In this study, we conducted a Monte Carlo simulation to compare the

performance of MNLFA and SEM trees and forests in detecting DIF and assessing MI under varying conditions. Specifically, we evaluate their effectiveness in identifying non-invariance and detecting relevant covariates. Our findings will inform best practices for selecting statistical techniques to test MI in psychological assessment.

Did any of the contributors already conduct related simulation studies on this specific question?

Neither of the authors has conducted simulation studies that are of immediate relevance to the current project before. However, Andreas Brandmaier was involved in conducting simulation studies on SEM trees/ SEM forests as well as SEM modeling in general (e.g., Buchberger et al. 2024; Díaz et al. 2025; Arnold et al. 2021).

Aims

What is the aim of the simulation study?

The aim of this simulation study is to evaluate different methods (i.e. MNLFA and SEM trees/ SEM forests) regarding their accuracy, power, and false-positive rate in detecting MI across different factor models (ADEMP category ‘hypothesis testing’).

Data-Generating Mechanism

How will the parameters for the data-generating mechanism (DGM) be specified?

The data will be generated parametrically. Different population structural equation models (SEM) with latent variables and continuous indicators will be simulated in two different simulation studies:

Study 1

Single-Factor Model with Moderated Parameters

We specify a confirmatory factor analysis model (model 1) with a single latent factor η and four observed indicators $\mathbf{x} = (x_1, \dots, x_4)^\top$. The influence of continuous moderator variables is introduced into selected parameters via predefined transformation functions

applied to underlying raw moderator variables.

The population model is:

$$\mathbf{x} = \boldsymbol{\nu}(Z) + \boldsymbol{\Lambda}_x(Z) \eta + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}_\varepsilon).$$

where

- $\boldsymbol{\nu}(Z)$ is the vector of intercepts,
- $\boldsymbol{\Lambda}_x(Z)$ is the 4×1 factor loading matrix,
- $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}_\varepsilon)$ is the residual vector,
- $\eta \sim \mathcal{N}(0, 1)$ is the latent factor,
- $Z = h(M)$ denotes the effective moderator entering the model equations, obtained by transforming one or more raw continuous moderator variables.

Moderator Transformation

Let $M \sim \mathcal{U}(-1, 1)$ denote a bounded uniformly distributed continuous covariate.

The effective moderator is defined as

$$Z = h(M),$$

where $h(\cdot)$ is chosen from the following deterministic transformation functions. Each transformation type is treated as a distinct condition in the simulation design:

1. Linear:

$$h(M) = M$$

In this case, the effective moderator is bounded to the interval $(-1, 1)$ and rotation symmetric around zero.

2. Sigmoid:

$$h(M) = a + \frac{b - a}{1 + \exp(-k(M - c))}, \quad a = -1, b = 1, c = 0, k = 10.$$

This transformation maps the input to the open interval $(-1, 1)$ and yields a bounded moderator, rotation symmetric around zero, exhibiting diminishing sensitivity at extreme values.

3. Quadratic:

$$h(M) = 2M^2 - 1$$

This transformation captures symmetric nonlinear moderation as a function of distance from the center while remaining bounded in $(-1, 1)$.

4. Noise:

$$h(M) = 0$$

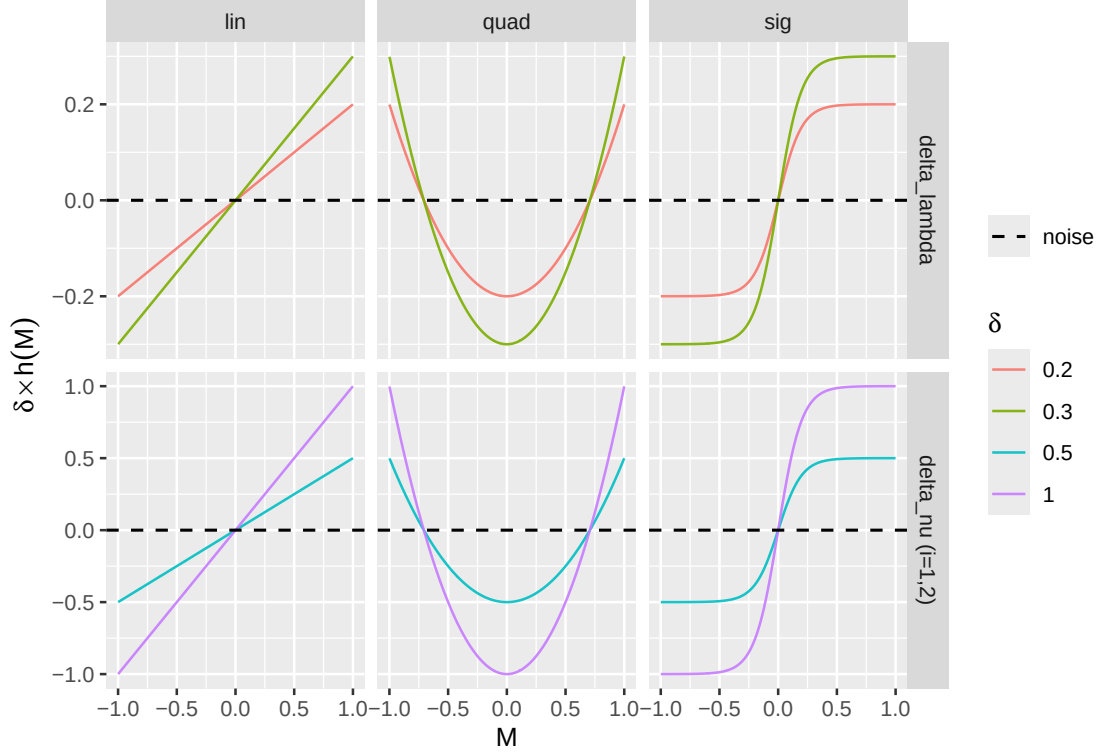
In this condition, the effective moderator is constant and therefore has no effect on any model parameter. Although this condition is termed “noise” in the simulation code, it represents the absence of systematic moderation rather than a randomly varying noise effect.

The absence of moderation is represented by the noise-moderator condition ($Z = 0$), not by setting $\delta = 0$; moderation coefficients are held fixed across conditions. All parameter-level moderation effects are specified as linear functions of the effective moderator Z . Nonlinearity in moderation arises exclusively through the transformation $Z = h(M)$, not through nonlinear parameter functions. The transformation $h(M)$ is part of the DGM in every moderation condition.

Factor Loadings

The baseline factor loadings were fixed to a discrete value of $\lambda_{xi} = 0.7$.

Variation in factor loadings across simulation conditions arises exclusively through continuous moderator effects specified in the population model.

**Figure 1**

Note. The dashed horizontal line indicates the null moderation case (noise: $\delta h(M) = 0$ for all M). The moderation strength for the latent covariance shares the same δ values as $\delta_\lambda \in \{0.2, 0.3\}$.

Moderated items

At the population level, the baseline (unmoderated) measurement model is specified with a single latent factor and four indicators. The factor loading matrix is constructed such that all four indicators share a common loading value λ :

$$\mathbf{\Lambda}_x = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix}$$

The latent factor variance is fixed at a prespecified constant value $\psi_\eta = 1$ and is held

constant across all simulation conditions.

For each condition involving moderation, the loading of item i is defined as a deterministic function of the moderator Z :

$$\lambda_{xi}(Z) = \lambda_{xi} + \delta_\lambda Z, \quad Z \in [-1, 1].$$

The parameter δ_λ represents the strength of the moderation effect and is specified using a discrete set of admissible values chosen to ensure that the moderated loading remains within a psychometrically plausible interval:

$$\lambda_{xi}(Z) \in [0.3, 1.0].$$

Given the fixed baseline loading $\lambda_{xi} = 0.7$, this constraint yields an admissible range

$$\delta_\lambda \in [-0.4, 0.3].$$

δ_λ is evaluated at the following discrete levels:

$$\delta_\lambda \in \{0.20, 0.30\}.$$

For moderator transformations beyond the linear case (quadratic or sigmoid), the functional form of $\lambda_{xi}(Z)$ is adapted accordingly, and the same discrete set of admissible moderation strengths is applied. This yields crossed combinations of (i) moderator transformation, (ii) moderated parameter type, (iii) number of moderated items, and (iv) moderation strength.

Factor loading matrix

$$\mathbf{\Lambda}_x(Z) = \begin{bmatrix} \lambda_1(Z) \\ \lambda_2(Z) \\ \lambda_3(Z) \\ \lambda_4(Z) \end{bmatrix}$$

Loading moderation is manipulated via the pattern of moderated items and the magnitude of the moderation effect. For each indicator i , the loading is modeled as

$$\lambda_i(Z) = \lambda_i + \delta_\lambda Z,$$

where λ_i denotes the baseline loading and δ_λ controls the strength and direction of loading moderation.

No loading moderation:

$$\delta_\lambda = 0 \quad \text{for all } i = 1, \dots, 4.$$

Partial loading moderation:

Only the loadings of items 1 and 2 vary with Z :

$$\delta_\lambda \in \{0.20, 0.30\} \quad \text{for } i \in \{1, 2\}, \quad \delta_\lambda = 0 \quad \text{for } i \in \{3, 4\}.$$

Full loading moderation:

All four loadings vary with Z :

$$\delta_\lambda \in \{-0.3, -0.2, 0.2, 0.3\} \quad \text{for } i = 1, \dots, 4.$$

Residual variances

Residual variances were chosen such that all four indicators had a reliability of

$$\rho = .75.$$

Given the fixed latent variance

$$\psi_\eta = 1$$

and baseline loading

$$\lambda_i = 0.70,$$

baseline residual variances were computed as

$$\theta_i = \frac{\lambda_i^2 \psi_\eta (1 - \rho)}{\rho}.$$

Residual variances were held constant across all simulation conditions. Consequently, when factor loadings were moderated, indicator reliability varied as a function of the moderator.

Intercepts

Intercept moderation is manipulated via the pattern of moderated items and the magnitude of the moderation effect. For each indicator i , the intercept is modeled as

$$\nu_i(Z) = \nu_i + \delta_\nu Z,$$

where ν_i denotes the baseline intercept and δ_ν controls the strength and direction of intercept moderation.

No intercept moderation:

$$\delta_\nu = 0 \quad \text{for all } i = 1, \dots, 4.$$

Partial intercept moderation:

Only the intercepts of items 1 and 2 vary with Z :

$$\delta_\nu \in \{0.5, 1.0\} \quad \text{for } i \in \{1, 2\}, \quad \delta_\nu = 0 \quad \text{for } i \in \{3, 4\}.$$

Full intercept moderation:

All four intercepts vary with Z :

$$\delta_\nu \in \{0.5, 1.0\} \quad \text{for } i = 1, \dots, 4.$$

Baseline intercepts are set to 1 across simulation conditions.

Population model overview

The population models are generated by combining the loading- and intercept-moderation specifications above.

Table 1

Population models defined by loading and intercept moderation specifications.

Model	Loading moderation	Intercept moderation	Active moderator structure
0	none	none	none
1.1	full	none	$Z_1 = h(M_1)$
1.11	none	full	$Z_1 = h(M_1)$
1.12	full	full	$Z_1 = h(M_1)$
1.2	partial, items 1–2	none	$Z_1 = h(M_1)$
1.21	none	partial, items 1–2	$Z_1 = h(M_1)$
1.22	partial, items 1–2	partial, items 1–2	$Z_1 = h(M_1)$
1.3	full	none	$Z_1 = h(M_1), Z_2 = h(M_2), Z_1 Z_2$
1.32	full	full	$Z_1 = h(M_1)$ (loadings), $Z_2 = h(M_2)$ (intercepts)

For Model 1.3, the loading equation is:

$$\lambda_i(Z_1, Z_2) = \lambda_i + \delta_\lambda Z_1 + \delta_\lambda Z_2 + \delta_\lambda^2 Z_1 Z_2$$

Specifically, the moderation coefficients are defined as

$$\delta_{\lambda 1} = \delta_{\lambda 2} = \delta_\lambda, \quad \delta_{\lambda 12} = \delta_\lambda^2.$$

For Model 1.32, loading and intercept moderation depend on separate moderators:

$$\lambda_i(Z_1) = \lambda_i + \delta_\lambda Z_1,$$

and

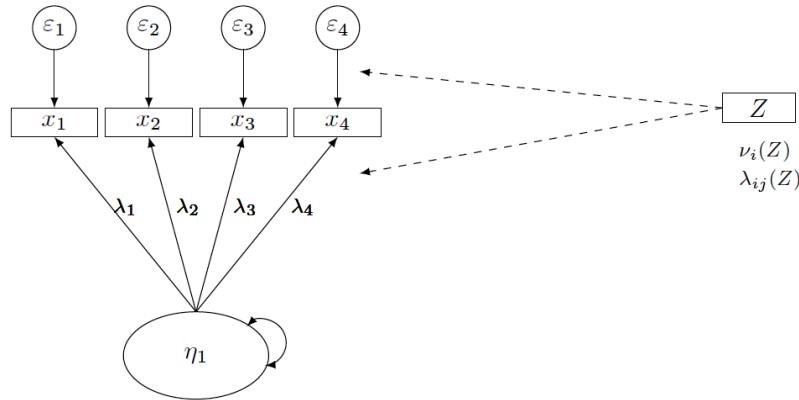
$$\nu_i(Z_2) = \nu_i + \delta_\nu Z_2.$$

Latent distribution

The latent factor is assumed to follow a standard normal distribution,

$$\eta \sim \mathcal{N}(0, 1),$$

so that the latent mean is fixed at 0 and the latent variance is fixed at 1 across all simulation conditions.

Analytical Model Study 1**Figure 2**

Note. Conceptual depiction of used Analytical Models.

The analytical model will contain one latent factor η_1 , with four manifest indicators $x_1, \dots, x_4$, their factor loadings $\lambda_1, \dots, \lambda_4$ and their respective residuals $\epsilon_1, \dots, \epsilon_4$ (See Figure 2). Residual variances are freely estimated, and residual covariances are fixed to zero.

For MNLFA, moderation of selected factor loadings and intercepts is modeled parametrically. Two analytical MNLFA specifications are considered. In the linear specification, measurement parameters are modeled as linear functions of the raw moderator M . In the quadratic specification, the moderator entering the MNLFA is transformed as $2M^2 - 1$. In the main study, two MNLFA specifications will be evaluated: one assuming linear moderation in the observed moderator(s), and one assuming quadratic moderation. Therefore, MNLFA is correctly specified only when the analytical functional form matches the DGM transformation $h(M)$, and is otherwise intentionally misspecified. For population models involving a single moderator, MNLFA includes the corresponding observed moderator as the predictor of measurement parameters. For population models involving a single moderator, MNLFA includes the corresponding observed moderator as the predictor of measurement parameters. For the interaction population model (Model 1.3), the analytical model additionally includes the interaction term $M_1 \times M_2$. For Model 1.32, separate observed

moderators (M_1 and M_2) are included for loading and intercept moderation, respectively.

What will be the different factors of the data-generating mechanism?

The simulation design includes several manipulated factors governing the data-generating mechanism (DGM). These factors are systematically crossed to generate the simulation conditions. The population models are summarized in Table 1.

Core DGM Factors

The following factors are systematically varied in the study using a grid search design:

Functional form of moderation. The functional form relating the raw moderator to the effective moderator is varied across the following deterministic transformations:

1. Linear
2. Quadratic
3. Sigmoid
4. Noise/null transformation, $h(M) = 0$

Strength of moderation. In each case where parameters are subjected to moderation, the strength of the moderation δ is varied:

δ_λ is evaluated at the following discrete levels:

$$\delta_\lambda \in \{0.2, 0.3\}.$$

whereas intercept moderation is evaluated using

$$\delta_\nu \in \{0.5, 1.0\}.$$

Population Model

Nine population models are considered, differing in the presence and pattern of loading moderation, intercept moderation, and moderator structure (Table 1).

Model misspecification of moderator form. The DGM always defines moderation through the transformed moderator $Z = h(M)$. Linear and quadratic MNLFA impose corresponding parametric moderation functions. Consequently, each MNLFA specification is correctly specified only when its assumed functional form matches the population model and is intentionally misspecified otherwise. SEM trees do not assume a parametric moderator function but instead identify MNI through recursive partitioning of the observed moderator variables.

Limitations of the Core DGM. The simulation assumes that the moderators are measured without error. Consequently, the results reflect an idealized setting in which moderation effects are perfectly observed. Introducing measurement error in the moderators would likely reduce statistical power and estimation accuracy but is beyond the scope of the present study.

Simulation Design Factors

The factorial simulation design crosses the following manipulated factors:

- Population model (9 levels; Table~1)
- Moderator transformation (linear, quadratic, sigmoid, noise/null)
- Loading moderation magnitude ($\delta_\lambda \in \{0.20, 0.30\}$)
- Intercept moderation magnitude ($\delta_\nu \in \{0.50, 1.00\}$)
- Sample size ($N \in \{250, 500, 1000\}$)
- Analysis method (linear MNLFA, quadratic MNLFA, SEM Trees)

If there is more than one factor: How will the factor levels be combined and how many simulation conditions will this create?

We employ a full grid search, in which parameter values are sampled from pre-specified discrete values. A total of 6,000 simulation replications should ideally be conducted. This number was chosen to ensure that the confidence intervals for key estimated proportions (e.g., power or Type I error rates near a binomial success probability of $p = 0.8$

achieve an approximate width of $\pm 1\%$. This is based on the calculation:

$$SE = 2 \times \sqrt{\frac{p(1-p)}{6000}},$$

which yields a standard error consistent with the desired precision for evaluating simulation outcomes within this study.

The implemented simulation employed a fully crossed factorial design. Crossing the nine population models, four moderator transformations, two loading-moderation magnitudes, two intercept-moderation magnitudes, and four sample sizes yielded 576 unique data-generating design cells. Each generated dataset was analyzed using three analytical methods (linear MNLFA, quadratic MNLFA, and SEM Trees), resulting in 1,728 method-specific analysis conditions. To ensure computational feasibility, each condition is replicated 100 times. In addition, one SEM Tree sensitivity condition evaluates the influence of ten irrelevant moderator variables.

Estimands and Targets

What will be the estimands and/or targets of the simulation study?

The primary estimands in this simulation study pertain to the detection and accurate estimation of MNI due to moderation effects on measurement parameters. Specifically, we assess whether the estimation methods compared can correctly identify the presence or absence of moderation in factor loadings and intercepts, reflecting the true data-generating structure. Consequently, key simulation outcomes include Type I error rates and statistical power associated with detecting such moderation effects.

Methods

How many and which methods will be included and which quantities will be extracted?

We will compare the following methods:

- 1) **Moderated Nonlinear Factor Analysis:** MNLFA was initially introduced by Bauer and Hussong (2009) within the framework of integrative data analysis and was subsequently extended into a general approach for assessing measurement invariance and moderation (Bauer 2017). The fundamental principle of MNLFA is the parameterization of both measurement and structural model parameters as functions of one or more moderator variables. MNLFA accommodates both frequentist estimation methods, such as maximum likelihood (ML), and Bayesian techniques, including Markov Chain Monte Carlo (MCMC), thereby offering methodological flexibility aligned with specific analytical objectives (Münch and Koch 2024).
- 2) **SEM Trees:** Structural equation modeling (SEM) trees were originally introduced by Brandmaier et al. (2013) as a way to systematically search for important covariates and their interactions in data, creating homogeneous subgroups. The method was refined later on by Brandmaier et al. (2016) and Arnold et al. (2021). SEM trees and forests are data-driven, non-parametric methods that build upon the decision tree paradigm, thus using recursive partitioning to identify latent subgroups in which model parameters differ, without assuming specific functional forms or predefined covariate effects. These approaches allow for nonlinear moderation of factor loadings and can reveal complex interaction effects, enabling the exploratory detection of DIF.

For the parametric MNLFA, we will extract estimated moderated parameters (e.g., factor loadings, intercepts), their associated standard errors, and two-sided p -values for testing the null hypothesis of no moderation effect. Additionally, model fit indices such as AIC and the LRT-statistics will be recorded.

For the parameter estimates and fit indices, we will summarize their distributions across simulation replications by reporting key quantiles (2.5th, 50th, and 97.5th percentiles). The null hypothesis will be rejected if the associated p -value is less than the conventional significance level of 0.05.

In contrast, non-parametric tree-based approaches do not yield direct parameter estimates with standard errors. We use score-based SEM-trees (Arnold et al. 2021). Their detection of MI/ non-invariance is operationalized via the presence of at least one split in the correct condition at the correct respective level of focus parameters. Those splits, introducing a subgroup, will be evaluated at the significance level: 0.05. Additionally, a Bonferroni correction to account for multiple comparisons at each node. The trees will be allowed to split on the focus parameters relevant for the invariance testing (the means of the intercepts, factor loadings and the latent covariance in study 2). Candidate subgroup structures are assessed by testing whether they produce statistically significant differences in the specified model parameters across groups. Differences in factor variances between subgroups are permitted in the model, but they are not used as criteria for selecting or evaluating covariates. Furthermore, pre-pruning will be applied, limiting the maximum depth of the tree to 3 and `min.N` is set to 50. Because the MNLFA also considers directly the form of the moderation, which the SEM trees do not, we further want to use SEM forests for generating partial dependence plots to get an approximation of the functional form of the moderating variables.

With this distinction we aim to ensure that each method is evaluated according to its own inferential framework, while maintaining comparability in terms of its ability to detect MI reliably.

Performance Measures

Which performance measures will be used?

For both the MNLFA approaches and SEM Trees, the primary performance measures concern the detection of MNI.

Type I Error Rate and Power: The rejection rate of the null hypothesis at significance level $\alpha = 0.05$, calculated as

$$\widehat{\text{RRate}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \mathbf{1}(p_i \leq 0.05),$$

where $\mathbf{1}(\cdot)$ is the indicator function.

Performance measures calculated for multiple moderated parameters (e.g., factor loadings or intercepts) will be summarized separately for loading and intercept moderation and, where appropriate, aggregated across parameters.

For the MNLFA approaches, parameter recovery will additionally be evaluated by comparing the estimated moderation parameters with their corresponding population values. The following classical Monte Carlo performance measures will be used:

Bias: The average difference between the estimated parameter $\hat{\theta}$ and the true parameter θ , defined as

$$\widehat{\text{Bias}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_i - \theta,$$

where n_{sim} is the number of simulation replications.

Absolute Bias:

$$\widehat{\text{AbsBias}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} |\hat{\theta}_i - \theta|.$$

Relative Bias: (expressed as a proportion of the true parameter)

$$\widehat{\text{RelBias}} = \frac{\widehat{\text{Bias}}}{|\theta|} \quad \text{for } \theta \neq 0.$$

Root Mean Squared Error (RMSE):

$$\widehat{\text{RMSE}} = \sqrt{\frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} (\hat{\theta}_i - \theta)^2}.$$

Coverage Probability: The proportion of confidence intervals that contain the true parameter θ .

How will Monte Carlo uncertainty of the estimated performance measures be calculated and reported?

Monte Carlo uncertainty will be quantified using Monte Carlo standard errors (MCSEs) for the estimated performance measures.

The **Rejection Rate** refers to the proportion of simulated datasets in which a null hypothesis is rejected, commonly used to estimate empirical Type I error rates or statistical power.

The **MCSE for the Rejection Rate** is calculated assuming a binomial distribution of rejections across n_{sim} simulation replications, which reflects the standard error of a proportion estimator:

$$\text{MCSE}_{\widehat{\text{RRate}}} = \sqrt{\frac{\widehat{\text{RRate}}(1 - \widehat{\text{RRate}})}{n_{\text{sim}}}},$$

For **Bias**, MCSE is calculated based on the sample variance of the specific parameter estimates across simulation replications:

$$\text{MCSE}_{\widehat{\text{Bias}}} = \frac{S_{\hat{\theta}}}{\sqrt{n_{\text{sim}}}},$$

where

$$S_{\hat{\theta}} = \sqrt{\sum_{i=1}^{n_{\text{sim}}} \{\hat{\theta}_i - (\sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_i / n_{\text{sim}})\}^2 / (n_{\text{sim}} - 1)}$$

is the sample standard deviation of the effect estimates.

Siepe et al. (2024), Morris et al. (2019) and Whitehead (1993)

How many simulation repetitions will be used for each condition?

The number of simulation repetitions and how we derived them is described above under “If there is more than one factor: How will the factor levels be combined and how

many simulation conditions will this create?”.

How will missing values due to non-convergence or other reasons be handled?

Non-convergence or other missing values may occur during model estimation. We plan to:

- Exclude only the specific simulation replicates that fail to converge for a given method while retaining data from other methods within the same replication.
- Report the proportion of non-converged cases per method and condition.
- If the proportion of non-convergence exceeds a pre-specified threshold (e.g., 5%), we will flag the corresponding simulation condition and interpret results with caution.

No imputation of missing estimates will be performed.

How do you plan on interpreting the performance measures?

The performance measures will be interpreted in terms of Type I error rates and statistical power. Type I error is defined as the proportion of replications in which a method incorrectly detects measurement noninvariance when none is present. Statistical power is defined as the proportion of replications in which a method correctly detects measurement noninvariance when it is truly present.

Results will be evaluated both overall and conditional on design factors. Particular attention will be paid to whether Type I error rates are controlled at nominal levels and how power varies as a function of effect size and data-generating conditions. Comparisons between methods will focus on their relative sensitivity to different forms of noninvariance (metric vs scalar) and robustness across conditions.

Other

Which statistical software/packages do you plan to use?

All simulations and analyses will be conducted in R (R Core Team 2025). The MNLFA-style models will be implemented directly in `OpenMx` (Boker et al. 2026), using

raw-data maximum likelihood estimation and definition-variable-style moderation through OpenMx algebra objects. SEM tree analyses will be performed with the `semtree` (Brandmaier et al. 2026) package using OpenMx RAM models. Simulation design construction, data handling, aggregation, and visualization will rely on `dplyr` (Wickham et al. 2026), `tidyr` (Wickham et al. 2025), `tibble` (Müller and Wickham 2026), `purrr` (Wickham and Henry 2026), `furrr` (Vaughan and Dancho 2022), `future` (Bengtsson 2021), `parallel` (R Core Team 2025), and `ggplot2` (Wickham 2016).

Which computational environment do you plan to use?

The simulation study will be conducted in R (R Core Team 2025).

Which other steps will you undertake to make simulation results reproducible?

We are planning to make the code of this project publicly available. Software and packages will be documented via `renv` (Ushey and Wickham 2026) and Docker. Furthermore, we are planning to use the `repro` package, as well as the `reproducibleRchunks` package for better reproducibility.

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